Variation of Parameters

can be used to for none-homogenus linear differential equations with constant coefficents with the right hand side can be anything

Steps

1) write the equation into standard form: $y'' + y' + y = f(x)$

2) solve the D.E for homogenus / set the D.E equal to zero find the y_c using the methods from homogenus linear equations with coefficents constant

3) compute the wronskian: $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

4) compute $u_1 = \int \frac{-y_2 f(x)}{W}$ and $u_2 = \int \frac{y_1 f(x)}{W}$

5) write the y_p using this equation: $y_p = u_1 y_1 + u_2 y_2$

6) write general equation: $y = y_c + y_p$

$y'' + y = \sin^2(x)$ — $f(x)$

• began by take this part and set it to zero to get y_c

$y'' + y = 0 \longrightarrow m^2 + 1 = 0 \longrightarrow m = \pm i$

• this is a case #3

$$y_c = C_1 \cos(x) + C_2 \sin(x)$$

• let y_1 and y_2 be any of the two terms in y_c

$y_1 = \cos(x) \qquad y_2 = \sin(x)$

• y_1 and y_2 can be any thing

• compute wronskian using y_1 and y_2

$$w = \begin{vmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{vmatrix} = \cos^2(x) - (-\sin^2(x)) = 1$$

• Plug into u_1 and u_2 equation and solve

$$u_1 = \int \frac{-y_2 f(x)}{w}$$

$$u_1 = -\int \frac{\sin(x) \cdot \sin^2(x)}{1} dx$$

$$u = -\int \sin^3(x) \, dx$$

- **Odd power of sine?** Save one sine → use $\sin^2 = 1 - \cos^2$.
- **Odd power of cosine?** Save one cosine → use $\cos^2 = 1 - \sin^2$.

$$u = -\int \sin(x) \left(1 - (\cos^2(x)\right)) \, dx$$

$$u = -\int \sin(x) - \sin(x) \cos^2(x) \, dx$$

$$u_2 = \int \frac{y_1 f(x)}{w}$$

$$u_2 = \int \frac{\cos(x) \sin^2(x)}{1} dx$$

$$u_2 = \int \cos(x) \sin^2(x) \, dx$$

$u = \sin(x)$

$du = \cos(x) \, dx$

$\frac{du}{\cos(x)} = dx$

$$u_2 = \int \cancel{\cos}(x) \cdot u^2 \cdot \frac{du}{\cancel{\cos}(x)}$$

$$u_2 = \int u^2 \, du$$

$$u_2 = \frac{(\sin(x))^3}{3}$$

$$- \left[\int \sin(x) - \int \sin(x)\cos^2(x) \right]$$

$$- \int \sin(x) + \int \sin(x)\cos^2(x)$$

$$-(-\cos(x)) +$$

$u = \cos(x)$

$du = -\sin(x)\, dx$

$$-\frac{du}{\sin(x)} = dx$$

$$-(-\cos(x)) - \int \cancel{\sin(x)}\, u^2 \cdot \frac{du}{\cancel{\sin(x)}}$$

$$u_1 = \cos(x) - \frac{(\cos(x))^3}{3}$$

- Plug into $y_p = u_1 y_1 + u_2 y_2$

$$y_p = \left(\cos(x) - \frac{(\cos(x))^3}{3}\right)(\cos(x)) + \left(\frac{(\sin(x))^3}{3}\right)(\sin(x))$$

$$y_p = \cos^2(x) - \frac{\cos^4(x)}{3} + \frac{\sin^4(x)}{3}$$

Plug into general equation: $y = y_c + y_p$

$$\boxed{y_c = C_1 \cos(x) + C_2 \sin(x) + \cos^2(x) - \frac{\cos^4(x)}{3} + \frac{\sin^4(x)}{3}}$$

$y'' + y = \csc(x) \leftarrow f(x)$

$y'' + y = 0 \longrightarrow m^2 + 1 = 0 \longrightarrow m = \pm i$ • this is case #3

$y_c = C_1 \cos(x) + C_2 \sin(x)$

$y_1 = \cos(x) \quad , \quad y_2 = \sin(x)$

$$W = \begin{vmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{vmatrix} = \cos^2(x) - (-\sin^2(x)) = 1$$

$u_1 = \int \frac{-y_2 f(x)}{W}$

$u_1 = \int \frac{-\sin(x)\csc(x)}{1} dx$

$u_1 = \int -\cancel{\sin(x)} \cdot \frac{1}{\cancel{\sin(x)}} dx$

$u_1 = \int -1 \, dx$

$u_1 = -x$

$u_2 = \int \frac{y_1 f(x)}{W}$

$u_2 = \int \cos(x) \cdot \csc(x) \, dx$

$u_2 = \int \cos(x) \cdot \frac{1}{\sin(x)} dx$

$u = \sin(x)$

$du = \cos(x) dx$

$\frac{du}{\cos(x)} = dx$

$u_2 = \int \cancel{\cos(x)} \cdot \frac{1}{u} \cdot \frac{du}{\cancel{\cos(x)}}$

$u_2 = \int \frac{1}{u} du$

$u_2 = \ln|\sin(x)|$

$y_p = u_1 y_1 + u_2 y_2$

$y_p = -x\cos(x) + \ln|\sin(x)| \sin(x)$

$$y = y_c + y_p$$

$$\boxed{y = C_1 \cos(x) + C_2 \sin(x) - x\cos(x) + \operatorname{Ln}|\sin(x)|\sin(x)}$$

$$y'' + 3y' + 2y = \frac{1}{3+e^x}$$

$$y'' + 3y' + 2y \longrightarrow m^2 + 3m + 2 = 0 \longrightarrow m = -1 \;,\; m = -2$$

• this is a case #1

$$y_c = C_1 e^{-x} + C_2 e^{-2x} \;,\quad y_1 = e^{-x} \;,\quad y_2 = e^{-2x}$$

$$W = \begin{vmatrix} e^{-x} & e^{-2x} \\ -e^{-x} & -2e^{-2x} \end{vmatrix} = (-2e^{-x} \cdot e^{-2x}) - (-e^{-x} \cdot e^{-2x})$$

$$= -e^{-3x}$$

$$u_1 = \int \frac{-y_2 f(x)}{W}$$

$$u_1 = \int \frac{-e^{-2x}\left(\frac{1}{3+e^x}\right)}{-e^{-3x}}$$

$$u_1 = \int \frac{e^x}{3+e^x}$$

$u = 3 + e^x$

$du = e^x \; dx$

$\frac{du}{e^x} = dx$

$$u_1 = \int \frac{\cancel{e^x}}{u} \cdot \frac{du}{\cancel{e^x}}$$

$$u_1 = \operatorname{LN}|3 + e^x|$$

$$u_2 = \int \frac{y_1 f(x)}{W}$$

$$u_2 = \int \frac{e^{-x}\left(\frac{1}{3+e^x}\right)}{e^{-3x}}$$

$$u_2 = -\int \frac{e^{2x}}{3+e^x}$$

$u = 3+e^x$

$du = e^x \; dx$

$\frac{du}{e^x} = dx$

$$u_2 = -\int \frac{e^x \cdot \cancel{e^x}}{u} \cdot \frac{du}{\cancel{e^x}}$$

$$u_2 = -\int \frac{e^x}{u} \, dx$$

$u = 3 + e^x$

$e^x = u - 3$

$$u_2 = -\int \frac{u-3}{u} \, dx$$

$$u_2 = -\int 1 - \frac{3}{u} \, du$$

$$u_2 = -(u - 3\ln u)$$

$$u_2 = -((3+e^x) - 3\ln|3+e^x|)$$

$$u_2 = -3 - e^x + 3\ln|3+e^x|$$

$$y_p = u_1 y_1 + u_2 y_2$$

$$y_p = (\ln|3+e^x|)(e^{-x}) + (-3 - e^x + 3\ln|3+e^x|)(e^{-2x})$$

$$\boxed{y = C_1 e^{-x} + C_2 e^{-2x} + e^{-x} (\ln|3+e^{-x}| - 1 + 3e^{-2x} (\ln(3+e^x) - 1)}$$